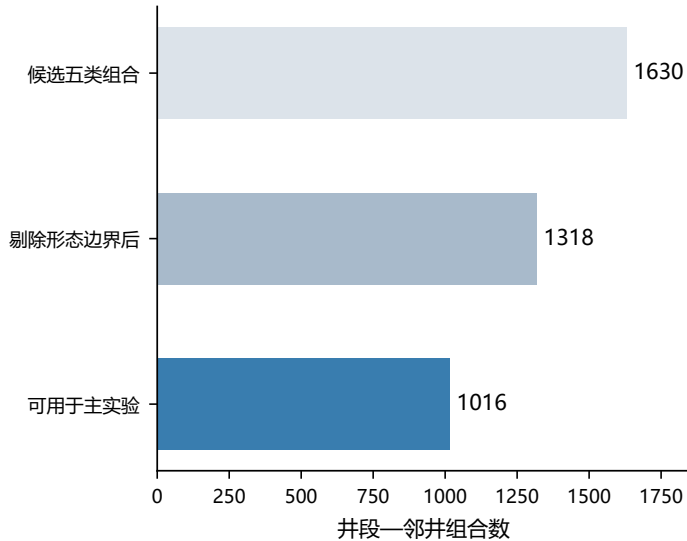
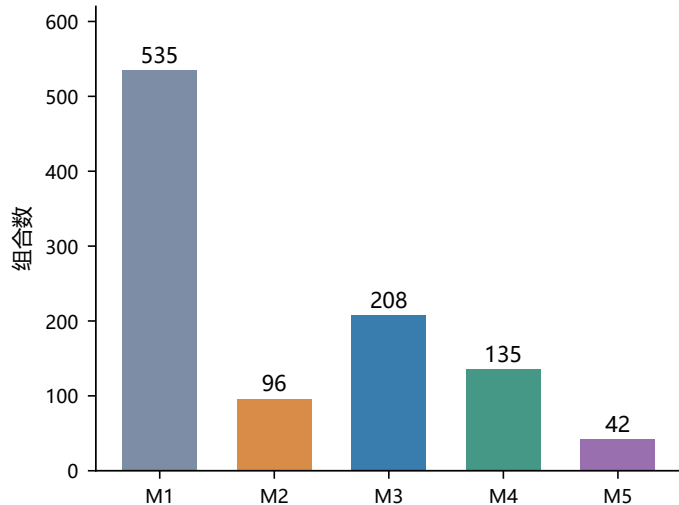


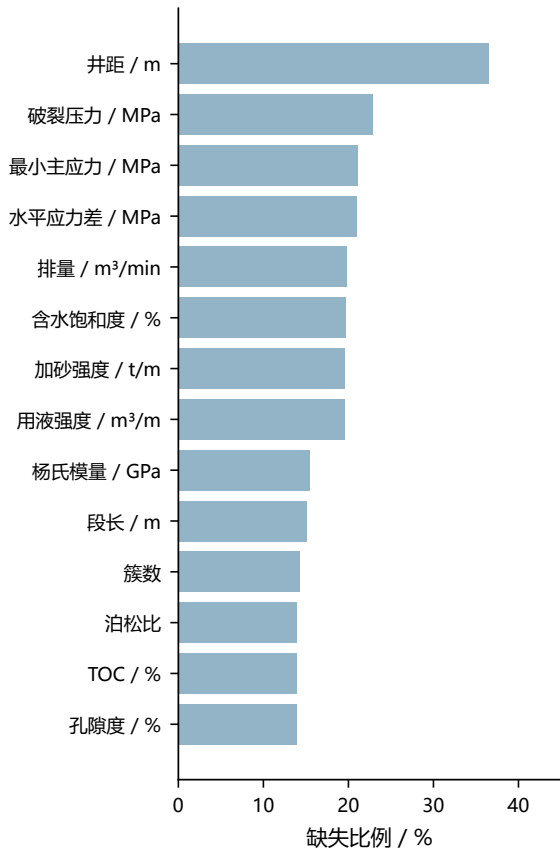
a 建模样本筛选



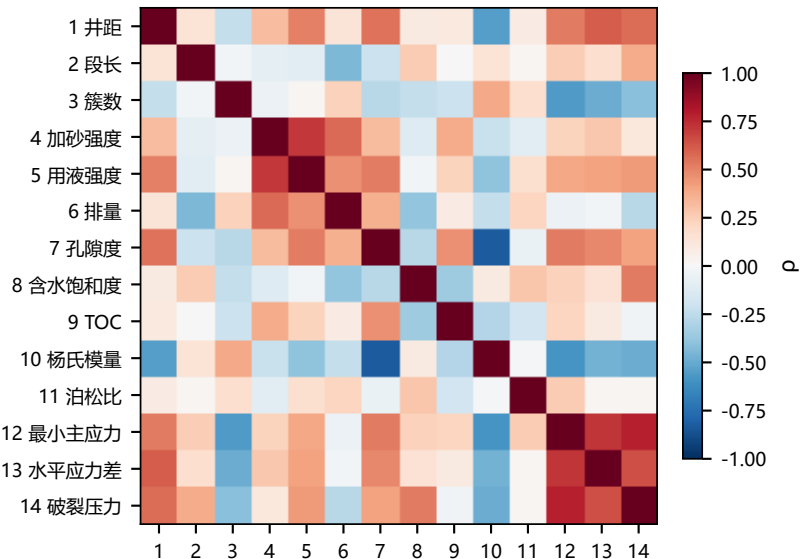
b 23个平台·21个独立分组



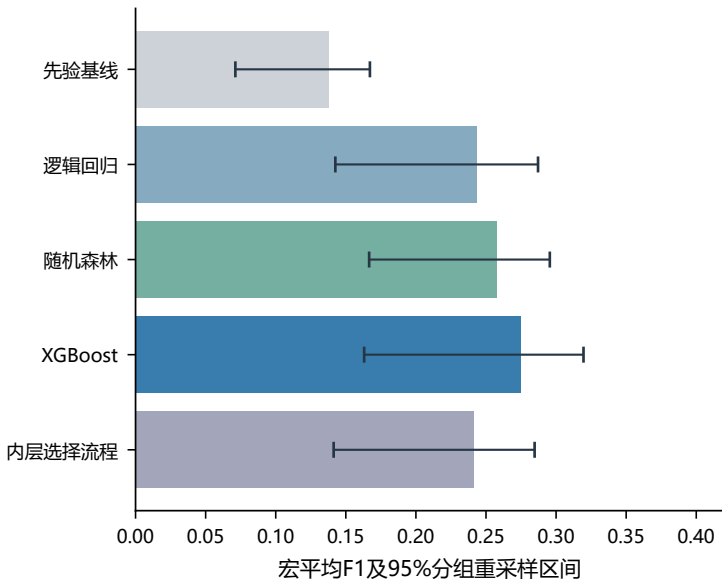
a 质控后的特征覆盖



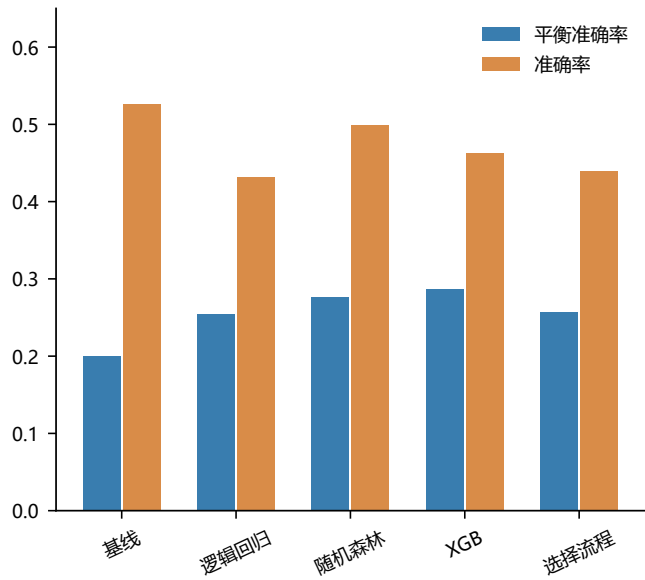
b Spearman相关系数



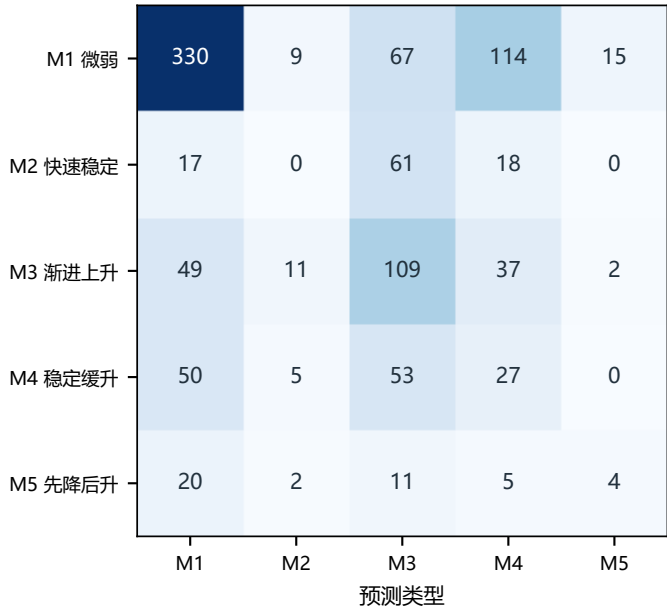
a 跨平台组折外表现



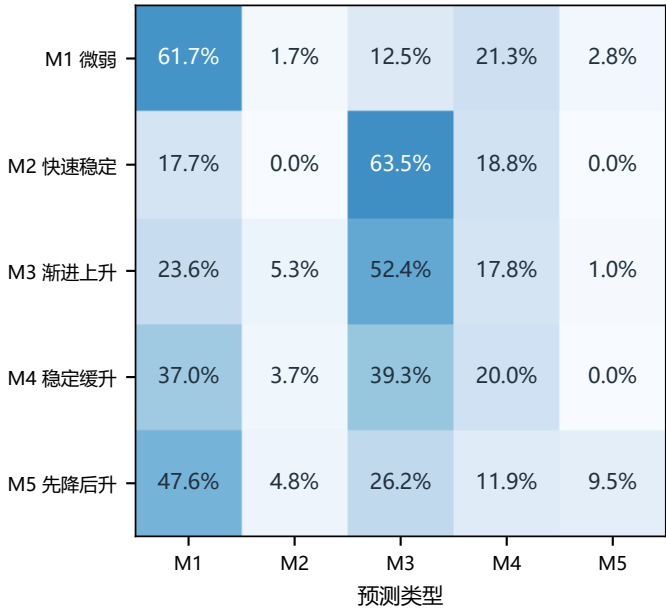
b 多数类影响下的不同指标



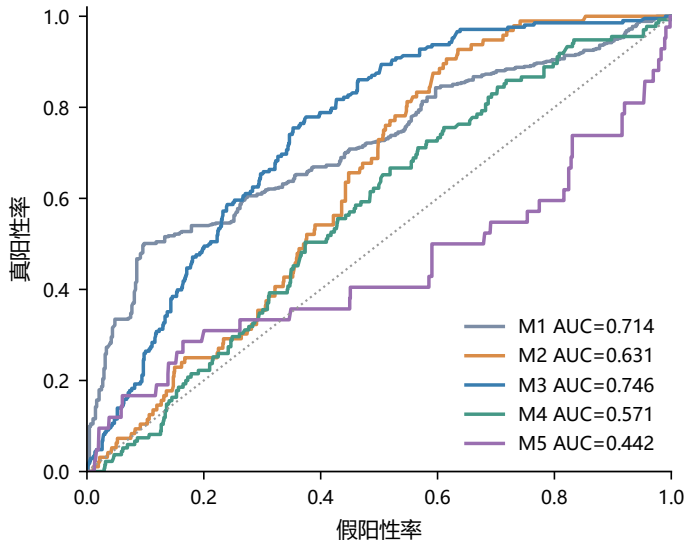
样本计数



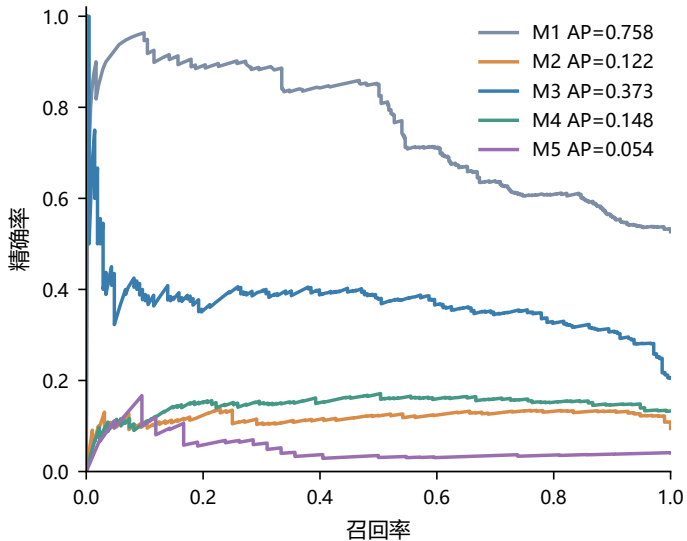
按真实类别归一化

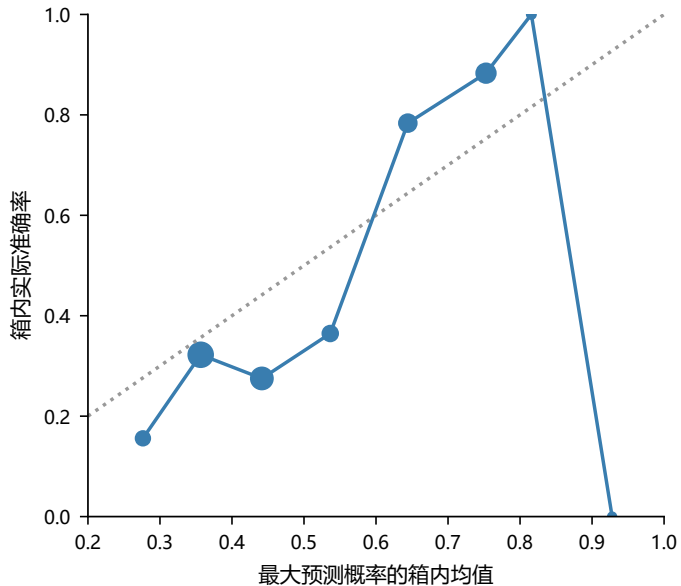
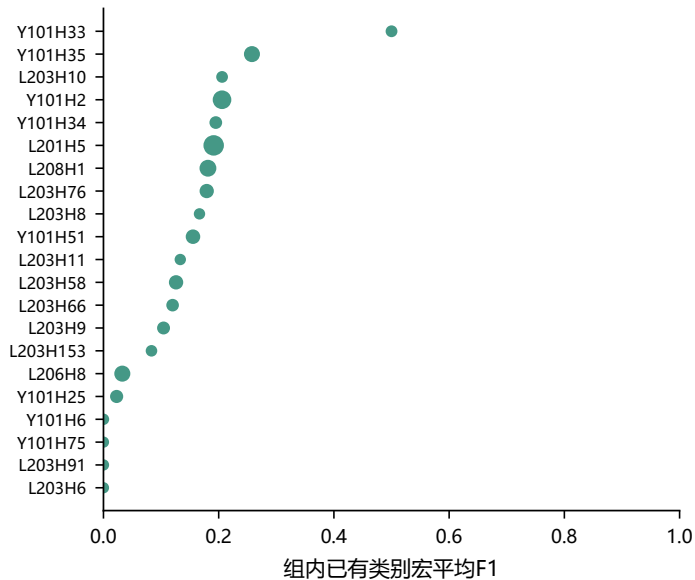


a 一对其余类别ROC

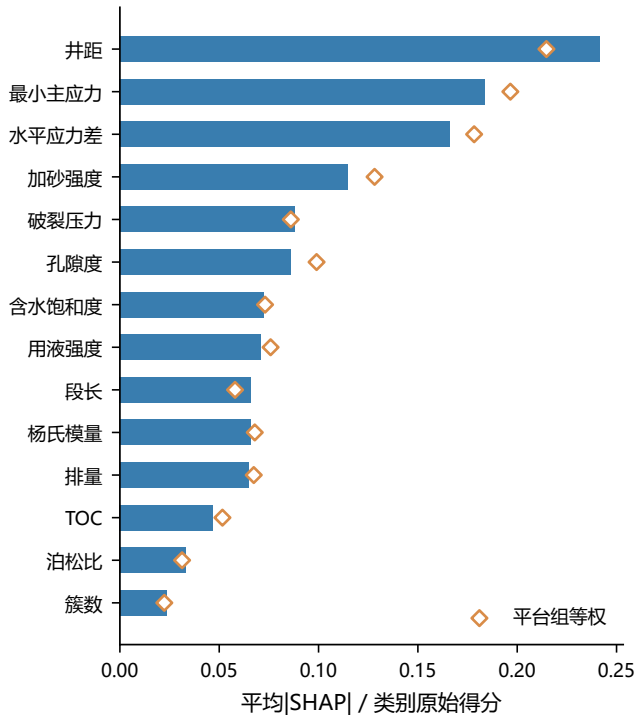


b 一对其余类别PR

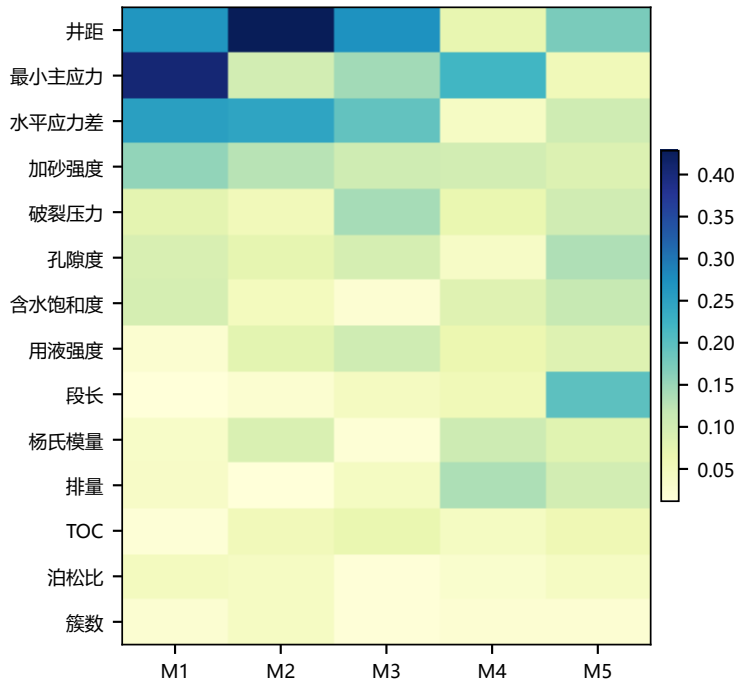


a 未校准概率可靠性**b 平台组间差异**

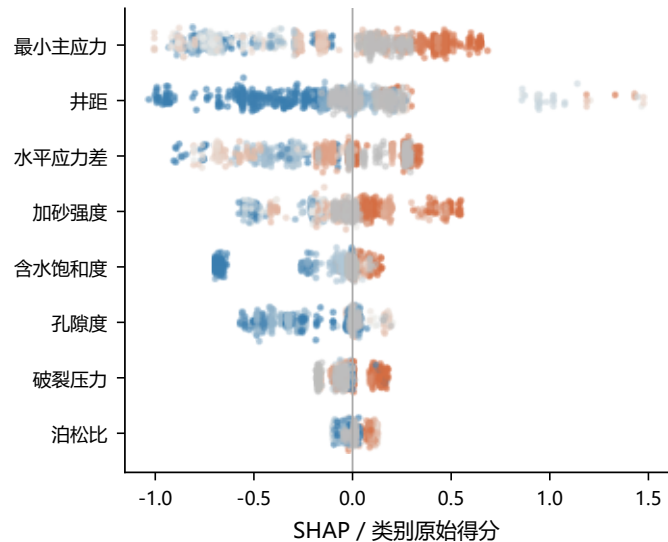
a XGBoost折外全局贡献



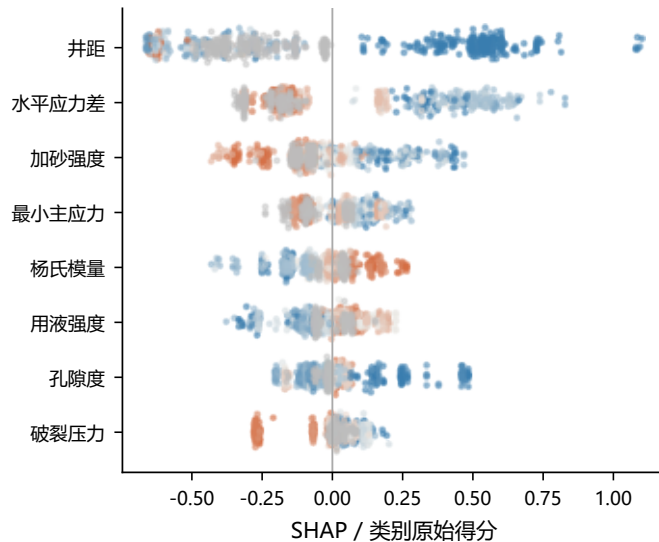
b 各类型平均绝对贡献



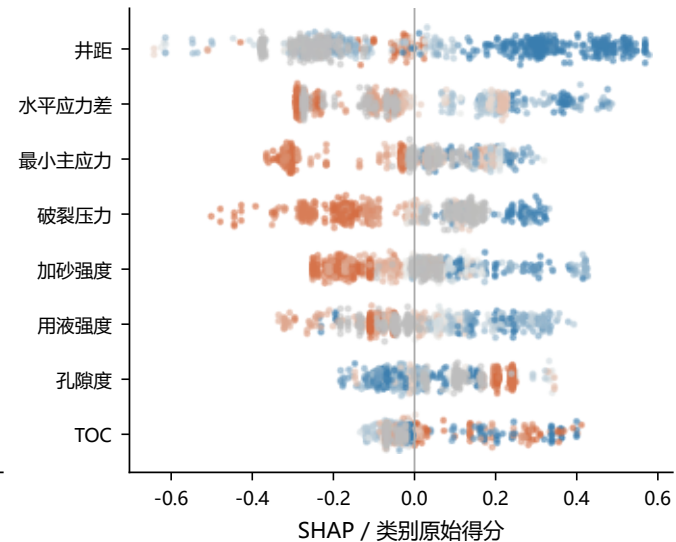
a M1 微弱



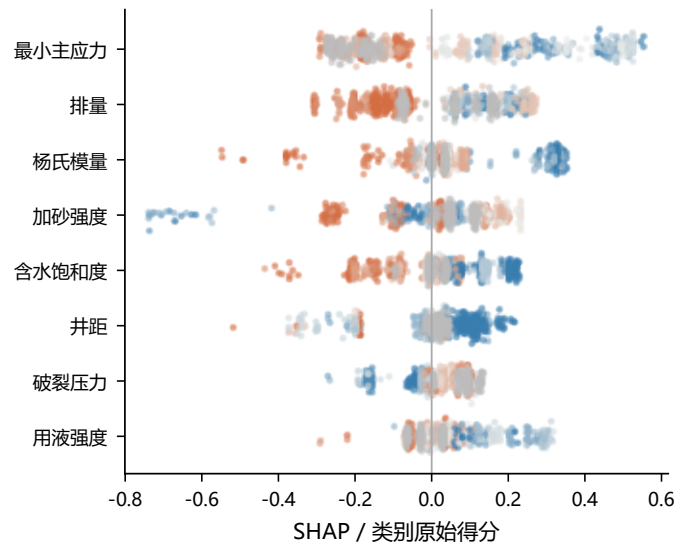
b M2 快速稳定



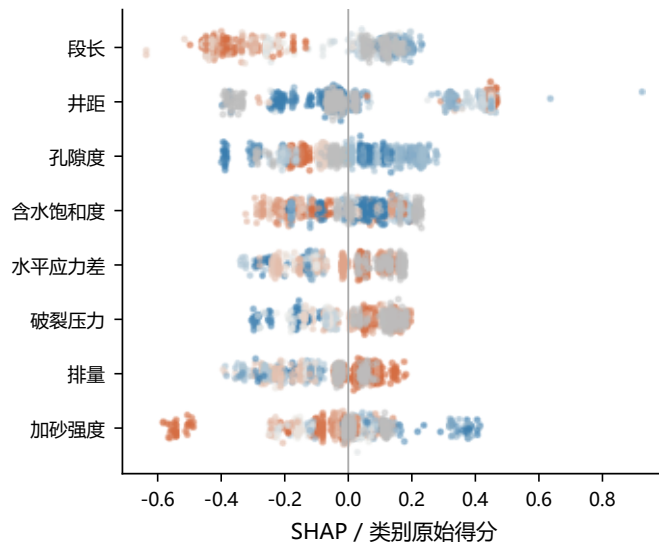
c M3 渐进上升



d M4 稳定缓升



e M5 先降后升

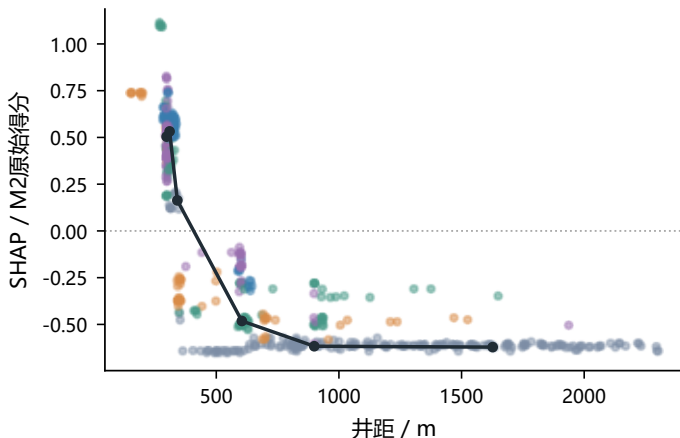


每点为一个折外样本
蓝色：特征值较低
橙色：特征值较高
灰色：原始值缺失

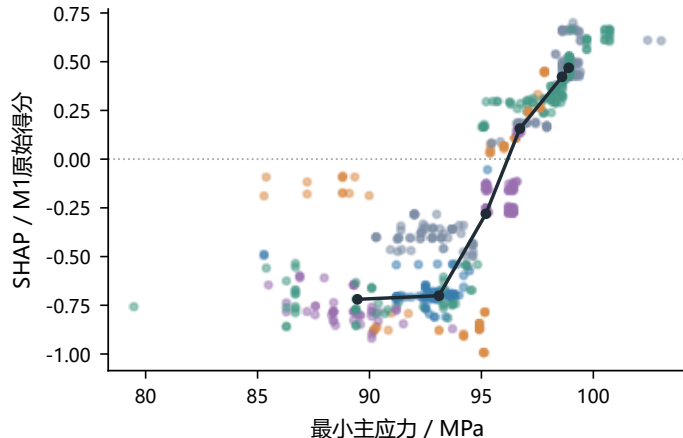
SHAP > 0 提高该类别原始得分
不代表风险增加或概率增量

● 验证折1 ● 验证折2 ● 验证折3 ● 验证折4 ● 验证折5

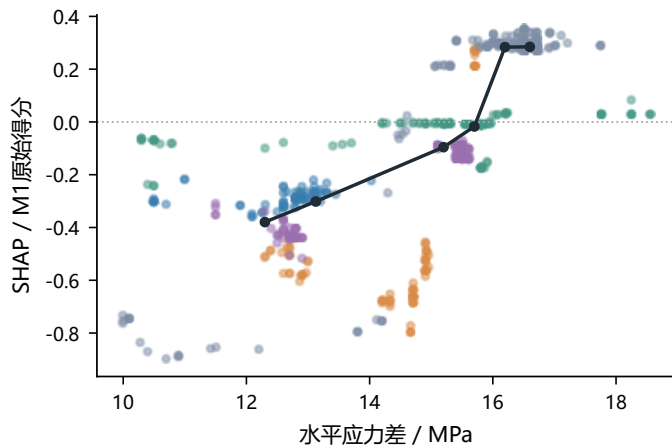
a 井距与快速稳定



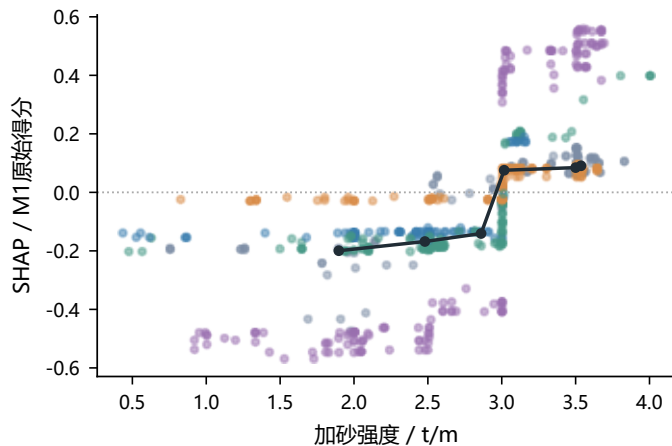
b 最小主应力与微弱



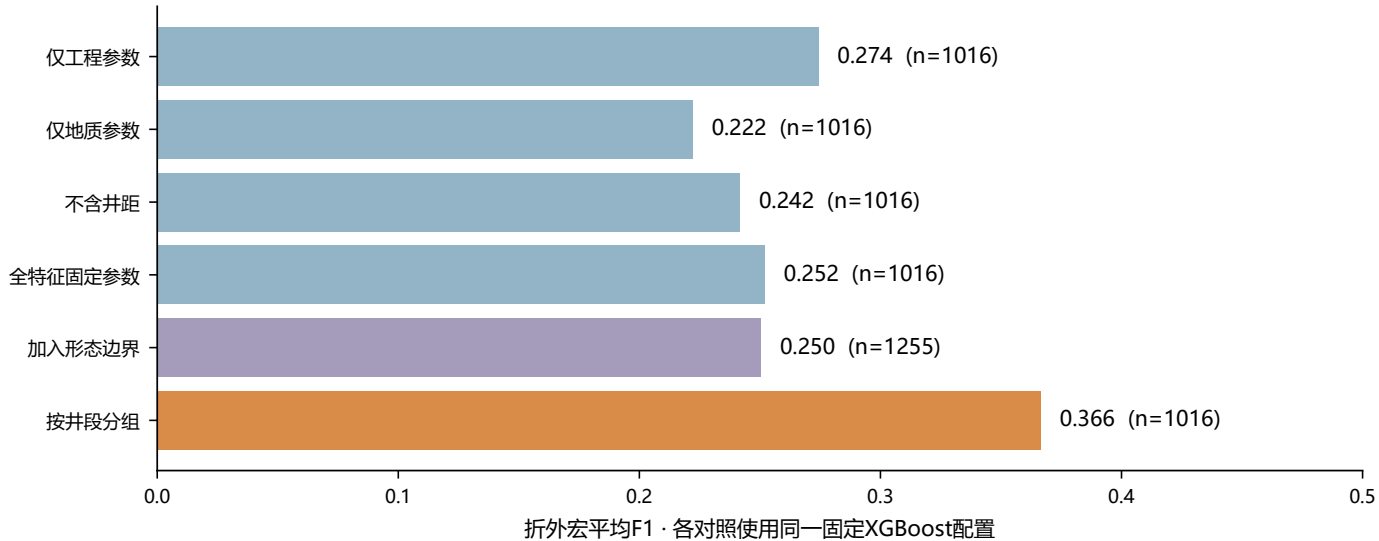
c 水平应力差与微弱



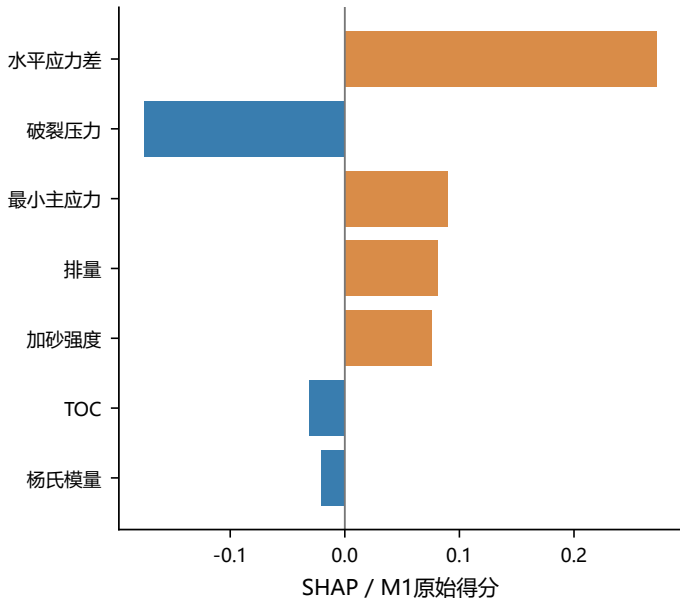
d 加砂强度与微弱



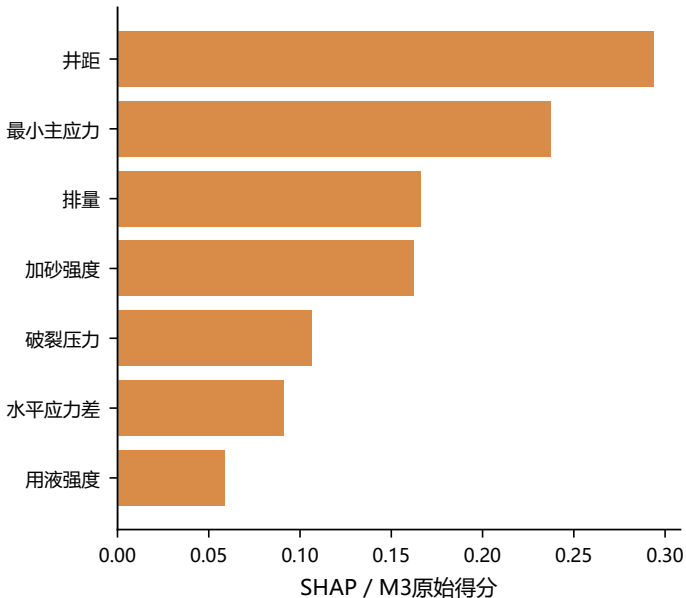
a 特征组及验证层级对照

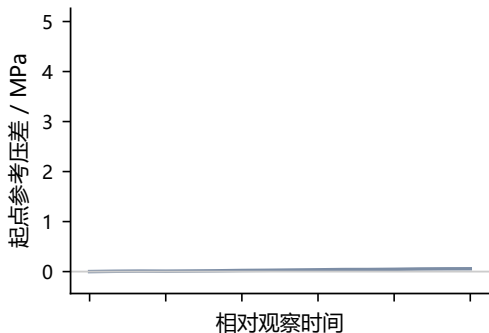
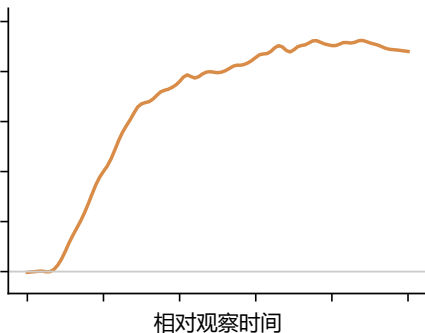
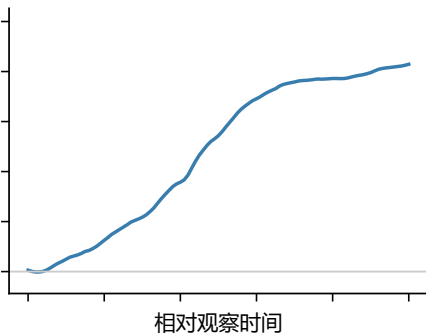
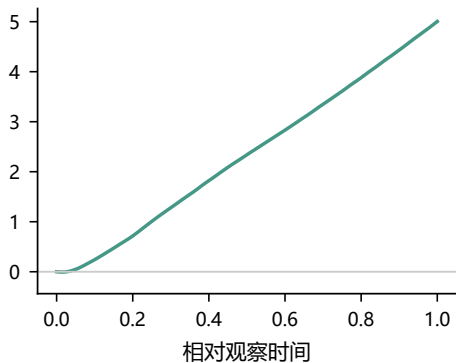
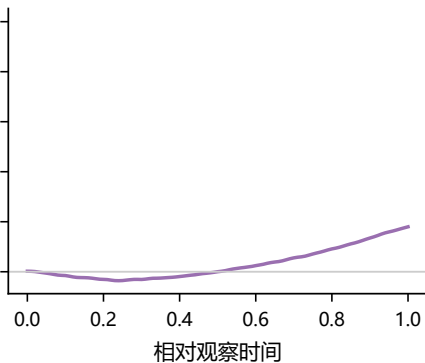


a 正确示例: M1 → M1



b 错误示例: M4 → M3



M1 微弱**M2 快速稳定****M3 渐进上升****M4 稳定缓升****M5 先降后升**

实际代表曲线
统一压差尺度
M1与M5幅度较小